

Designation: B 1567-24

Standard Specification for Titanium Grade 5 Alloy Bar

This standard was issued on March 22, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers titanium Grade 5 (Ti-6Al-4V) alloy bars for general engineering and aerospace applications

1.2 The titanium alloy bars covered by this specification are intended for use in manufacturing of aerospace components, medical implants, and high-performance automotive parts requiring excellent strength-to-weight ratio and corrosion resistance.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

B 348 Specification for Titanium and Titanium Alloy Bars and Billets

E 8 Test Methods for Tension Testing of Metallic Materials

E 384 Test Method for Microindentation Hardness of Materials

E 1409 Test Method for Determination of Oxygen and Nitrogen in Titanium

3. Terminology

3.1 Definitions:

3.1.1 alpha-beta alloy—a titanium alloy containing both alpha and beta phase stabilizing elements

3.1.2 mill annealed—heat treatment condition produced during manufacturing to optimize mechanical properties

3.1.3 ultimate tensile strength—the maximum stress that a material can withstand while being stretched

4. Classification

4.1 titanium alloy bars shall be classified according to diameter range and heat treatment condition.

4.1.1 **Grade 5A**—Annealed condition, diameter 6-25 mm

4.1.2 **Grade 5B**—Annealed condition, diameter 25-75 mm

4.1.3 **Grade 5C**—Solution treated and aged, diameter 6-50 mm

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Grade (5A, 5B, or 5C)

5.1.3 Diameter (mm)

5.1.4 Length (m)

5.1.5 Quantity (number of bars or total weight in kg)

5.1.6 Heat treatment condition

6. Materials and Manufacture

6.1 The titanium alloy shall be produced by vacuum arc remelting (VAR) or vacuum induction melting (VIM).

6.2 The material shall be hot worked and heat treated to meet the specified mechanical properties.

6.3 Chemical composition shall conform to requirements specified in Section 7.

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Aluminum (Al)	5.5 - 6.75%
Vanadium (V)	3.5 - 4.5%
Iron (Fe), max	0.30%
Oxygen (O), max	0.20%
Carbon (C), max	0.08%
Nitrogen (N), max	0.05%
Hydrogen (H), max	0.015%
Titanium (Ti)	Balance

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Tensile strength	≥ 895 MPa	ASTM E 8
Yield strength	≥ 828 MPa	ASTM E 8
Elongation	$\geq 10\%$	ASTM E 8
Hardness	280-340 HV	ASTM E 384

9. Dimensions and Permissible Variations

9.1 Bar diameter tolerance shall be ± 0.5 mm for diameters up to 25 mm.

9.2 For diameters greater than 25 mm, tolerance shall be ± 1.0 mm.

9.3 Length tolerance shall be +25 mm/-0 mm.

9.4 Straightness shall not exceed 2 mm per meter of length.

10. Workmanship, Finish, and Appearance

10.1 Bars shall be free from cracks, laps, seams, and other surface defects. Minor surface imperfections may be removed by grinding provided the diameter remains within tolerance.

11. Sampling

11.1 One tensile specimen and one chemistry sample shall be taken from each heat. Additional testing may be required for critical applications.

12. Test Methods

12.1 **Tensile Testing**—Machine tensile specimens per ASTM E 8 and test at ambient temperature.

12.2 **Chemical Analysis**—Perform chemical analysis using appropriate methods for each element.

12.3 **Hardness Testing**—Conduct Vickers hardness testing per ASTM E 384 on prepared cross-sections.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Each bar shall be tagged or marked with heat number, grade designation, and manufacturer identification.

17. Packaging and Package Marking

17.1 Bars shall be bundled and wrapped to prevent damage and contamination during transit. Protective end caps shall be applied.

18. Keywords

18.1 titanium; Ti-6Al-4V; Grade 5; bar; aerospace; alloy; biocompatible